The Application of Ant Colony Optimization Algorithm in Linear-Combination Blind Source Separation Problem

Nian Zhang
1.Institute of Geophysics and Geomatics,China University of Geosciences, Wuhan 430074 , China
2. Department of Computer Science and Information Technology,Guangzhou Maritime College,Guangzhou 510725,China

Tianyou Liu
Institute of Geophysics and Geomatics,China University of Geosciences, Wuhan 430074 , China

***Abstract*—The paper establishes a mathematical model of the Blind Source Separation (BSS),and introduce the basic theory of Ant Colony Optimization(ACO) algorithm. According to the idea of Independent component analysis (ICA),the paper proposes a linear blind source separation algorithm based on ant colony optimization.the algorithm establishes a cost function based on Minimum Mutual Information (MMI).The paper does the global optimization in cost function by using the the global optimization capability,positive feedback and other characteristics of ACO,and solves a class of BSS problem.The simulation experiment verifies the effectiveness of the algorithm.**

Keywords- Blind Source Separation; Ant Colony Optimization; Independent component analysis

I. INTRODUCTION

Blind Source Separation (BSS) is a focus of the modern signal processing,many scholars made some researches on the theory and algorithm of this issue for its widely application prospect on many fields such as wireless communication, image processing,speech signal processing,earthquake signal processing,array signal processing ,biomedical signal processing and so on, and they got attractive research results.

Ant colony optimization is a simulated evolutionary heuristic algorithm proposed by Italian scholar Dorigo M and his team in 1991. As a new intelligent optimization algorithm, ACO conquers the defect that Gradient algorithm falls into a local optimum easily,it is used widely in the solution of Portfolio optimization. By applying ACO to the solution of BSS problem.we can get better performance and more practical value in the BSS algorithm.

II. BLIND SOURCE SEPARATION

A. The Model of BSS

BSS means the process of recovering the source signal from the observed signal merely by the statistical properties of source signal in the condition that source signal and transmission channel are unknown. This paper only discusses the linear-combination BSS.We presume that there are n source signals, $S(t)=[s_1(t),s_2(t), \cdots , s_n(t)]^T$.m observed signals, $X(t)=[x_1(t),x_2(t), \cdots,x_m(t)]^T$, then we can describe the BSS by following equation:

$$X(t)=AS(t).$$

Among them ,A is m×n dimensions mixed matrix. Since C.Jutten and J.Herault proposed the famous H-J algorithm in 1991,many scholars make researches on the BSS and propose various optimize algorithm,their forms may be different,but the model is basically identical. Such as Figure 1,it is the basic model of BSS.

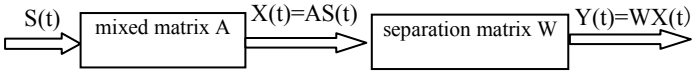


Figure 1. the base model of BSS

Among them, A is the mixing matrix, W is the separation matrix, the meaning of this model is: in the case that A is unknown and the weight is excepted independent, we know nothing about s(t),we need to find the separated matrix W to make each weight in the separated result Y(t)=WX(t) keep independent and close to s(t) as far as possible.

B. Hypothesis of Model

Due to the lack of the structural information of mixing matrix,it may lead to the multiple solutions of equation,so we need to do the following assumption.

(1)The component of Source signal S(t) is statistical independence between each other,and the number of the weights that obey Gaussian distribution will not be more than one.

(2)All source signals $S_i(t)$ are 0 average, stationary stochastic process is unit variance.

(3)The number of observed signals is no less than the number of source signals.

C. Independent Component Analysis

Independent component analysis is one of the main algorithms to solve the BSS issue,the basic idea is that we can

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minimize the interdependence between components of different signals and make the mutual statistical independent as far as possible by transforming signals. To solve the BSS by applying ICA,it is Generally completed in two steps:whiten,to solve separation matrix.

Solving the separation matrix W is also composed of two parts:cost function,optimization algorithm. Choose different cost functions we can get different ICA algorithm,but optimization algorithm needs fast convergence speed and good stability. Cost function is foundation to solve the BSS. Currently,the cost function we commonly used are: maximum entropy,maximum negative entropy, minimum mutual information, the fourth-order cumulant, Kullbak-Leibler divergence etc.There are many optimization algorithms also,such as genetic algorithm,gradient algorithm and so on.This paper takes mutual information as cost function.

III. ANT COLONY OPTIMIZATION ALGORITHM

A. Algorithm Overview

Ant colony optimization algorithm is inspired from the foraging behavior of ants. In the process of Ant's movement,for finding the path Ant releases the chemical material called pheromone in the path passed by. When ant meets the intersection that never walked before,it selects a road randomly,at the same time,it releases the pheromone with the information about the length of the path. The longer the path is, the fewer pheromones released. When the ant which comes later meets the intersection again,it will choose the path with more pheromone (namely the path with shorter distance) to forward, thus form a positive feedback mechanism. The pheromone on the optimal path gets more and more, and the pheromone on other paths get less and less until it gradually disappear. finally,all the ants will find the optimal path.

B. Algorithm Principle

Such as graph 2,it is a schematic diagram of the ACO system.

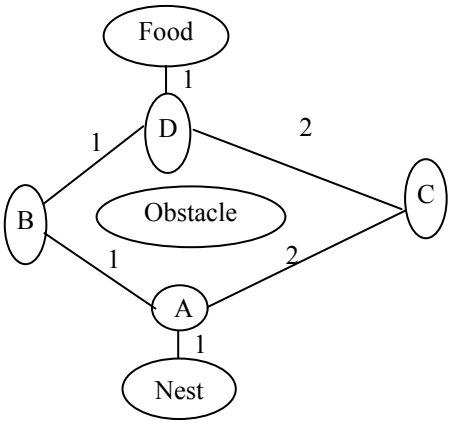


Figure 2 schematic diagram of the ACO system

As it can be seen,there are two paths from the ant-nest to food source: Nest-ABD-Food and Nest-ACD-Food, Respectively,the length is 4 and 6. Initially, there is no pheromone on two paths,the ants will choose a path randomly in a equal probability,assuming every ant has the same rate, because the Nest-ABD-Food is shorter than Nest-ACD-Food, the ant following the path of Nest-ABD-Food will reach the food source in a shorter time. So Nest-ABD-Food will be deposited with more pheromone within the same time.

The more pheromone,the more probability of the path chosen by the following ants,take such a cycle of feedback, ultimately,most ants will choose the shorter path Nest-ABD-Food.

IV. THE APPLICATION OF ACO ALGORITHM TO BSS

A. The cost function based on MMI

From the knowledge of information theory, we know that mutual information is a measurement used to measure the independence between random variables. The mutual information is always non-negative,if and only if the random variable is statistical independence between each other,the mutual information is zero. Minimum Mutual Information (MMI) derives the iterative formula of separation matrix W through minimizing the mutual information of Y directly which is outputted by system.

Actually,this kind of mutual information is the correlation between the compenent of Y which is determined by Kullback-Leibler (K-L) divergence I(W) between joint probability density function p (y; W) of Y and the product of marginal probability density function $\overline{p}$ (y;W):

$$I(Y,W)=D(p(y;W)\| \overline{p}(y;W))$$
$$=\int p(y;W)\log(p(y;W)/\overline{p}(y;W))dy$$

Among them, $\overline{p}(y;W)=\prod_{i=1}^{n} p_i(y_i;W)$ means the marginal probability density function product of output signal Y. $p_i(y_i;W)=\int p(y;W)dy_1\ldots dy_j\ldots dy_n, j=1,\ldots,n, j\neq i$, represents marginal probability density function of Y.

Through analysis and conversion, the above cost function can be simplified as:

$$J(Y;W)=\sum_{i=1}^{n} H(yi;W)-\frac{1}{2}\log|\det(WW^T)|-H(V_i^T X)$$

By using Gram-Charlier series expension of probability density to approach each output weight's edge pdf,we can get the cost function of over-determined and positive definite problem such as (1), (2) follows:

$$\min J(Y;W)=\frac{1}{2}\log\left|\det(WW^T)\right|+\frac{n}{2}\ln(2\pi e)-\sum_{i=1}^{n}[\frac{(k_3^i)^2}{2\cdot 3!}+\frac{(k_4^i)^2}{2\cdot 4!}-\frac{5}{8}(k_3^i)^2k_4^i-\frac{1}{16}(k_4^i)^3] \quad \ldots\ldots\ldots(1)$$

$$\min J(Y;W)=\log\left|\det(W)\right|+\frac{n}{2}\ln(2\pi e)-\sum_{i=1}^{n}[\frac{(k_3^i)^2}{2\cdot 3!}+\frac{(k_4^i)^2}{2\cdot 4!}-\frac{5}{8}(k_3^i)^2k_4^i-\frac{1}{16}(k_4^i)^3] \quad \ldots\ldots\ldots (2)$$

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Among them, $k_3^i = m_3^i$, $k_4^i = m_4^i - 3$, $k_k^i = E[(y_i)^k]$ is the k-order moment of y_i, then we optimize (1), (2)type by using ACO algorithm, and get the separation matrix W to make each component of Y tend to be independent.

B. Solve ICA Model by Using ACO Algorithm

1) Basic idea

The basic idea of this paper is that we can do the intelligent searching on the separation matrix by using ACO algorithm. First, the range of separation matrix W is set, for facilitate description, the range of each weight of W can be set as: $-1 \leqslant wy \leqslant 1$,the range is divided into several regions, the ant colony do the global optimization in the range of value; Second, each region can be divided into several sub-regional dynamically, to ensure that the ant colony do the local searching.

2) Solution Steps

Firstly, whiten observation signals X(t), then search separation matrix W by using ACO algorithm, the concrete steps are as follows:

(1) Initialized ACO, we can get a ant colony which is composed by m×n ants，put ant in each interval randomly, and initialize pheromone.

(2) Divide the interval chosen by ants into a number of sub-regional dynamically, to facilitate ants search optimization locally.

(3) The individual choice in ant colony has the greatest probability interval for the transfer, respectively, the formula is (3),(4):

Ant moved from t time interval g to the t+1 time interval h，the corresponding separation matrix update from W(t) to W(t+1).

$$h=\arg\max_{h\in allowed(t+1)}\{P_{g,h}\} \quad \text{.....(3)}$$

Among them， $P_{g,h}$ is the probability transfer from interval g to h. $\arg\max_{h\in allowed(t+1)}$ represents the interval corresponding to the biggest transfer probability; $h \in$ allowed(t+1) represents the collection of interval at t+1 time, then the number of ant colony search interval can be limited effectively. The calculation formula of transfer probability $P_{g,h}$ is:

$$P_{g,h}=\begin{cases} \tau_h^{\alpha}\,\eta_{g,h}^{\beta} & h\in \text{allowed (t+1)} \\ 0, & h \in \text{allowed (t+1)} \end{cases} \quad \text{.....(4)}$$

Among them, τ_h is the pheromone of interval h， $\eta_{g,h}$ is the inspiration value of pheromone in ants' search of optimal interval, $\eta_{g,h}=F(J_g, J_h)$; $F(J_g, J_h)$ is the expectation that transfer from g interval to h interval; J means the cost function value of each interval, we can see its formula from the aboved type (1),(2); Non-negative parameters α , β means information accumulated in the movement of information and the different roles that inspiration factor plays in ants choosing intervals.

(4)After global and local optimization,we update global pheromone according to the rules of type (5)，(6),toward to the the first step continuing cycle, carry out the optimization of M generation ant colony.

$$\tau_i \leftarrow \rho\,\tau_i + \triangle\tau_i^k \quad \text{.....(5)}$$

$$\triangle\tau_i^k = D\times\frac{1}{J_i} \quad \text{.....(6)}$$

Among them, $0 \leq \rho < 1$,ρ represents the durability of the pheromone material, 1-ρ represents the passage extent of the pheromone material.

After completing these steps aboved,combine with the convergence criteria of Cost function, judge the optimization results of M generation ant colony, Select a reasonable separation matrix corresponding to a generation of ant colony, isolated source signal from mixing signal.

C. Arithmetic Simulation

Such as figure 3(a), they are waveforms of three source signals S_1、S_2、S_3.

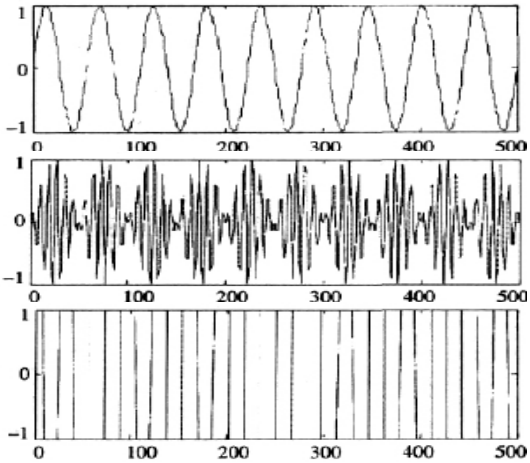
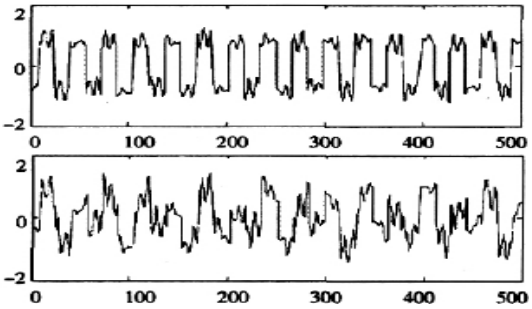


Figure3(a) source signals

Ignore the noise, and linear mix the source signal by using mixed matrix, we can get the mixed signal as Figure3(b).



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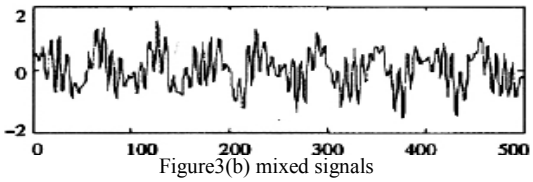


Figure3(b) mixed signals

Optimize the minimum mutual information cost function by using the ACO algorithm mentioned above, we can get the estimation of source signal, namely separation signals, their waveform is shown as figure 3(c).

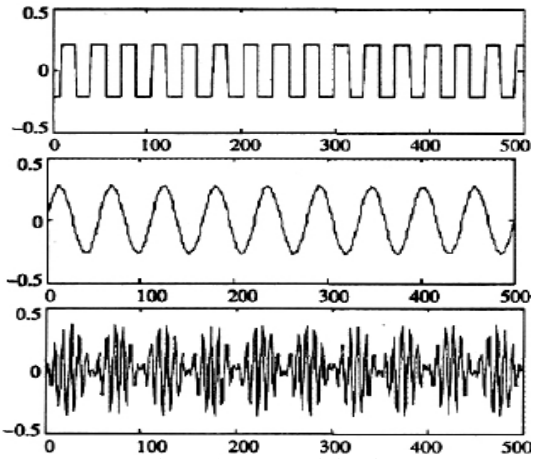


Figure3(c) separated signals

It can be seen that the separated signal (figure 3(c)) and the source signal(figure 3(a)) are very similar except their orders, and the separation effect is satisfactory. We need to say that this algorithm is applicable to blind separation of low dimensional only; to the high dimensional, it seems very tedious to optimize by using ACO algorithm.

V. CONCLUSIONS AND PROSPECT

Global optimization the cost function by using the ability of global optimization、positive feedback、strong robust etc characters of ACO algorithm, we can avoid the defect of local optimum of gradient method .Experiments show that the linear ICA algorithm based on ACO can solve overdetermination BSS problem and the mix of sub-gaussian and super-gaussian signals BSS problem.

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